



KLE Technological University

Creating Value,
Leveraging Knowledge

Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

DSP Report
on
MUSIC SOURCE SEPARATION

By:

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Under the Guidance of

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ENGINEERING

CERTIFICATE

This is to certify that project entitled “Music Source Seperation” is a bonafide work carried out by the student team of “Nandeesh katkol-02FE22BEC040, Priyanshu Porwal-02FE22BEC056, Jamuna Pulpatli-02FE22BEC032, Priyanka Budihal-02FE22BEC054”. The project report has been approved as it satisfies the requirements with respect to the dsp project work prescribed by the university curriculum for B.E. (V Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2024-2025.

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ABSTRACT

This project investigates a DSP-based system designed for vocal extraction from audio tracks. The method provides an efficient, real-time solution without reliance on machine learning, making it ideal for applications like karaoke, remixing, and music production using Stereo channel subtraction, the system suppresses centered vocal components by leveraging differences in dual-channel audio. Median filtering further refines the separation by reducing noise and residual artifacts. The approach stands out due to its computational efficiency, real-time processing capability, and wide applicability. Despite its advantages, the system assumes perfectly centered vocals, which may not hold for all tracks, and may struggle with complex audio mixtures. However, this method offers significant potential in audio engineering, with scope for enhancement to address its limitations.

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Chapter 1

Introduction

Vocal removal, or extracting vocals from music tracks, plays a vital role in applications like remixing, karaoke, and audio analysis. This process enables users to isolate vocals or background music, allowing flexibility in music production and research. Conventional approaches relied on filters and frequency separation, which often struggled with intricate audio mixtures. However, stereo channel subtraction offers a simpler yet effective method by leveraging the natural differences between left and right channels in dual-channel audio. In this approach, the difference between stereo channels is computed to suppress the centered vocal components while preserving the background music. Median filtering is then applied to enhance the separation quality further. This DSP-based method avoids reliance on machine learning algorithms and focuses entirely on signal processing techniques, making it an efficient solution for vocal extraction and karaoke track creation. The approach is widely applicable in remixing, music production, and other audio engineering tasks.

1.1 Problem Statement

To design a DSP-based system that extracts vocals from an audio track. This approach aims to create an efficient and accessible method for vocal isolation suitable for real-time applications such as karaoke, source separation, etc.

1.2 Background

Vocal removal, or extracting vocals from music tracks, plays a vital role in applications like remixing, karaoke, and audio analysis. This process enables users to isolate vocals or background music, allowing flexibility in music production and research.

Conventional approaches relied on filters and frequency separation, which often struggled with intricate audio mixtures. However, stereo channel subtraction offers a simpler yet effective method by leveraging the natural differences between left and right channels in dual-channel audio. In this approach, the difference between stereo channels is computed to suppress the centered vocal components while preserving the background music. Median filtering is then applied to enhance the separation quality further.

This DSP-based method avoids reliance on machine learning algorithms and focuses entirely on signal processing techniques, making it an efficient solution for vocal extraction and karaoke track creation. The approach is widely applicable in remixing, music production, and other audio engineering tasks.

1.3 Objectives

1.Real-Time Application

Ensure the system is capable of performing vocal isolation in real-time, making it suitable for applications like karaoke and live performances.

2.Ensure Wide Applicability

Enable the system to be used in various scenarios, such as music production, remixing, and audio analysis.

3.Develop a DSP-Based System

Create an efficient digital signal processing (DSP) system for extracting vocals from audio tracks.

Chapter 2

Literature Survey

1.Multichannel signal separation has been a key focus in audio processing, particularly for isolating vocals from background music. Yellin and Weinstein (1996) explored methods leveraging multichannel inputs to achieve blind source separation, laying the groundwork for various DSP techniques. In this project, stereo channel subtraction is applied to suppress vocals, followed by a band stop IIR filter targeting vocal frequencies (100 to 3000 Hz). This approach aligns with the principles outlined in their work, demonstrating the effectiveness of combining channel separation with frequency domain filtering for enhanced audio extraction.

2.Singer melody extraction from polyphonic signals involves developing a system to isolate and extract the singer's melody from music containing multiple sources. This process often employs blind source separation techniques. Key methodologies include the Short-Time Fourier Transform (STFT) for frequency domain analysis, Gaussian Mixture Models (GMMs) for statistical modeling, and the source/filter model for representing the singer's voice. Non-Negative Matrix Factorization (NMF) is another significant technique used for decomposing audio signals into meaningful components. However, GMMs have limited capability to handle complex characteristics of music, such as overlapping notes and multiple sound sources, necessitating advanced approaches for effective melody extraction.

3.CycleGAN-based audio source separation using time-frequency masking is a proposed method that leverages CycleGANs to generate and refine the separation of audio sources. The approach incorporates a multipitch tracking algorithm to estimate pitch tracks and utilizes binary masks for time-frequency filtering, enhancing the separation process. Although this method shows promise, it may require substantial computational resources for training the GANs. Additionally, the complexity of the approach may limit its suitability for real-time applications.

4.Blind Source Separation (BSS) and Independent Component Analysis (ICA) are essential techniques in signal processing. ICA is a statistical method that improves principal component analysis (PCA) by focusing on higher-order statistics, such as kurtosis, to achieve source separation. This involves utilizing a de-mixing matrix to maximize the statistical independence of the separated sources.ICA requires the sources to be statistically independent for successful separation. However, the method may encounter challenges when the sources are weakly independent or when noise interference is present, making source separation less effective in such conditions.

5.Harmonic/Percussive Separation using Median Filtering is a technique applied on spectrograms to enhance harmonic or percussive components by emphasizing horizontal or vertical elements. This process involves mask generation for separation, utilizing binary and soft masks. While effective in many cases, some percussive traces may remain in the harmonic output and vice versa, limiting its performance. Additionally, the method may not be suitable for real-time applications without further optimization.

Chapter 3

Methodology

Stereo audio tracks typically position vocals at the center of the mix, while instruments and other elements are distributed across the stereo field, creating a distinct spatial arrangement. Leveraging this characteristic, the system employs a channel subtraction technique, where one stereo channel (left) is subtracted from the other (right). This process effectively diminishes centrally positioned components, such as vocals, while retaining the stereo elements associated with the background music. This technique serves as a robust foundation for isolating vocal and instrumental components

3.1 Flow Chart

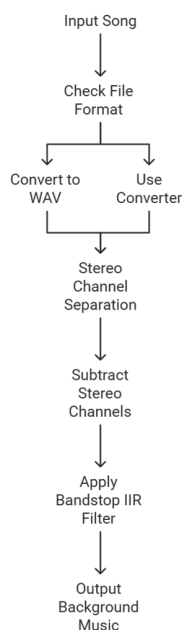


Figure 3.1: Flow Chart

3.2 Audio File Input

- The system primarily processes audio files in .wav format, which ensures compatibility and high-quality audio for precise analysis.
- If the input is in .mp3 format, a built-in conversion feature enables seamless transformation to .wav format using MATLAB's `audioread()` and `audiowrite()` functions. This ensures uniformity in file handling and processing.

3.3 Preprocessing

- Input File Validation: Ensure the input file is in .wav format. If not, convert .mp3 files to .wav.
- Channel Separation: Separate the stereo channels (left and right) for further processing.

3.4 Frequency Analysis

- Transform the separated signals into the frequency domain using `fft()`.
- Examine the frequency components of each separated source to identify the vocals and instruments.

3.5 Stereo Subtraction

-Stereo Channel Subtraction: The stereo audio signal is processed by subtracting the left channel from the right channel (or vice versa). This technique exploits the spatial arrangement of vocals (typically centered in the mix) to isolate them from the background music.

-Vocal Isolation: The subtraction results in an attenuation of the centrally-positioned vocals, leaving behind primarily the background music and instrumental components.

3.6 Filtering and Signal Enhancement

- Band-Stop Filtering: A band-stop filter is designed using MATLAB's `designfilt()` function. This filter specifically targets the vocal frequency range (300 Hz to 3000 Hz) to suppress residual vocals in the background music after the stereo channel subtraction.

-Signal Refinement: After applying the band-stop filter to the background music, the instrumental track is enhanced by attenuating the frequencies where vocals are most prominent, resulting in a cleaner and more isolated instrumental output.

3.7 Output Generation

-Saving Separated Tracks: After the separation process, the instrumental and vocal tracks are saved to disk using MATLAB's `audiowrite()` function. The instrumental track (after vocal suppression) and the vocal track (filtered signal) are saved as separate audio files in .wav format.

-Playback Verification: The quality of the separation is verified through playback of the instrumental and vocal tracks. The user can listen to both tracks to ensure the vocals are sufficiently removed from the background music.

-Spectrogram Analysis: Optionally, a spectrogram analysis can be performed to visually verify the separation quality. By plotting the spectrogram of the original and separated signals, the user can assess how well the vocals have been isolated and whether any unwanted artifacts remain.

3.8 Limitations

1. Separation Accuracy: The stereo channel subtraction method assumes that vocals are centrally panned. This approach may fail in cases where vocals are panned to one side or mixed unusually.

2. Stereo Configuration Dependency: The method is optimized for standard stereo audio mixes. Tracks in mono or with non-traditional stereo setups may not yield optimal separation results.

3. Frequency Overlap Challenges: Vocals and instrumental elements often occupy overlapping frequency ranges, which complicates the separation process and may degrade the instrumental quality.

4. Computational Demands: The stereo channel subtraction and filtering process can be computationally demanding, especially for large or high-resolution audio files, requiring significant processing power and memory.

Chapter 4

Result and Discussion

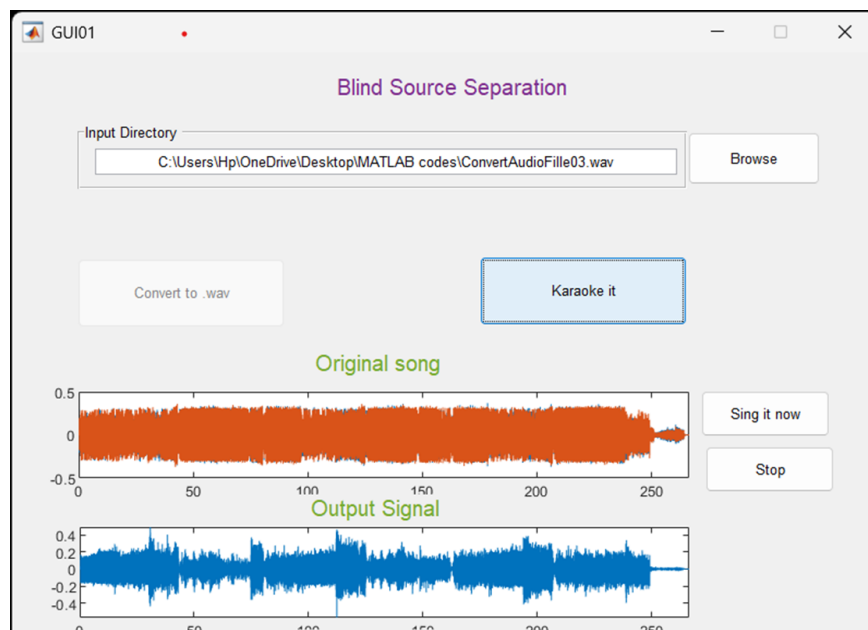


Figure 4.1: Result

The system provides an intuitive MATLAB GUI for karaoke extraction, allowing users to select audio files, convert them to .wav format if needed, and process them to generate the original song and extracted background music plots. The core technique involves stereo channel subtraction, where the left and right channels are used to attenuate the vocals, preserving the background music.

The system effectively isolates vocals in most standard stereo mixes, but residual vocal artifacts remain in cases of heavy reverb or complex mixing due to overlapping frequencies. A band-stop filter is used to further suppress these residual vocals, but its performance depends on the input's quality. The separation quality is assessed visually via waveform plots. While the method works well for moderate-length tracks, real-time processing is computationally challenging.

4.1 System Analysis

The karaoke extraction system uses stereo channel subtraction to isolate instrumental tracks by attenuating centered vocals. Its effectiveness depends on the stereo mix quality and frequency overlap between vocals and instruments. Limitations include residual vocals, potential quality loss, and computational demands for longer or high-resolution tracks. Future improvements could focus on refining subtraction techniques and exploring machine learning for better separation accuracy.

Chapter 5

Applications

1. **Cost-Effective Karaoke Creation:** The system provides an affordable solution for creating karaoke tracks without the need for expensive studio setups or specialized software.
2. **Easy Music Separation:** It allows efficient separation of vocals and instrumental tracks, ideal for remixing, karaoke performances, or music production.
3. **Versatile Format Support:** The system supports standard stereo audio formats (.wav, .mp3), ensuring compatibility across a wide range of music genres.
4. **User-Friendly Interface:** With the MATLAB-based GUI, the process is simple and accessible, requiring minimal technical expertise for users to operate effectively.

Chapter 6

Advantages

1. Cost-Effective Karaoke Creation: It enables the generation of karaoke tracks without requiring professional studio equipment.
2. Flexibility in Music Use: Vocals and instrumental tracks can be easily separated, making it useful for remixing, cover performances, and music production.
3. Wide Compatibility: The system works with standard stereo audio files, making it accessible for a variety of music genres and formats.
4. Ease of Use: The methodology uses familiar audio processing techniques, making it straightforward for users with basic MATLAB knowledge to apply.

Chapter 7

Future work

Future improvements to this DSP-based karaoke extraction project could focus on enhancing the accuracy of vocal isolation through advanced algorithms, such as adaptive filtering or machine learning-based techniques. The system could also be adapted to handle mono tracks or unconventional mixing by refining frequency-domain analysis and filtering methods.

A key area of development would be enabling real-time processing for live karaoke applications. Additionally, expanding the system's capability to handle various music genres and improving noise reduction techniques would further enhance audio quality. Future versions could also prioritize the creation of more intuitive, user-friendly interfaces to make the system accessible to those without advanced signal processing knowledge.

Chapter 8

Conclusion

This project effectively demonstrates the application of DSP techniques to separate vocals from instrumental tracks. By utilizing harmonic features and frequency analysis, the system successfully isolates vocals, making it suitable for remixing, cover creation, and music analysis. Unlike machine learning methods, the approach relies on traditional DSP, ensuring accessibility and efficiency for users with basic signal processing knowledge. With further refinements, such as enhanced filtering techniques, improved frequency analysis, and addressing residual artifacts, the system can achieve more precise separation for complex audio mixes. Additionally, incorporating noise reduction and multi-genre support would expand its usability. This DSP-based solution offers a cost-effective and practical tool for creating high-quality karaoke tracks, benefiting musicians, content creators, and audio enthusiasts.

Chapter 9

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